

U.S. Department of Homeland Security
U.S. Customs and Border Protection



Attachment 01a – Functional Requirements Document

For:

U.S. CUSTOMS AND BORDER PROTECTION
LARGE-SCALE FIXED
NON-INTRUSIVE INSPECTION SYSTEMS

Multi-Energy Portal Systems

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January 30, 2026
Version 2.6

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Revision Summary

The following *Table 1: Revision Table* indicates the changes made to the document since the last release.

Table 1 Revision Table

Version	Date	Location	Description
1.0	8/14/2019	Washington D.C.	Initial Draft
1.1	1/23/2020	Washington D.C.	Edits from vendor response to draft RFP
1.2	4/30/2020	Washington D.C.	Edits – SOW reconciliation
1.3	8/10/2020	Washington D.C.	Final Edits
2.0	7/20/2025	Washington D.C.	Updated with changes from lessons learned
2.1	7/25/2025	Washington D.C.	Comments and suggested revisions made by NII Procurement Team
2.2	8/22/2025	Washington D.C.	Updated based on FRD/SOW Crosswalk and Lessons Learned
2.3	09/24/2025	Washington D.C	Updated based on Stakeholder Comments
2.4	10/14/2025	Washington D.C	Additional Systems Engineering Comments
2.5	12/18/2025	Washington, D.C	Adjudicated RFI Responses
2.6	1/29/2026	Washington, D.C	Additional TID comments

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1 Introduction

United States Customs and Border Protection (CBP) Office of Field Operations (OFO) facilitates legitimate trade and travel into and out of the U.S. CBP Officers must screen all incoming persons and inspect vehicles, conveyances and cargo to detect and resolve potential threats. CBP Officers stand ready to interdict terrorists and their weapons, illicit narcotics and numerous forms of contraband smuggled into the United States. CBP Officers also detect and intercept foreign nationals illegally attempting entry into the United States. In addition, CBP Officers safeguard United States agriculture from harmful animal, fruit, and vegetable diseases and foreign invasive pest species not currently in the United States.

The Multi-Energy Portal (MEP) system offers a variety of transmission and backscatter X-ray imaging capabilities/energies which allow safe scanning of occupied vehicle cabs (per ANSI 43.17 2009) and higher flux multi-energy scanning of cargo trailers and sea containers, even through dense cargo or shielding. Multi-energy transmission X-ray allows the MEP to discern varying cargo densities which can support detection of anomalies within the cargo load. MEP configurations can also include under-vehicle and multi-view backscatter imaging.

The MEP systems include ancillary system components to assist in identifying the vehicle load or conveyance. It includes either license plate reader (LPR), Quick Response (QR) code readers and Radio Frequency Identification (RFID) systems for land border applications and Optical Character Recognition (OCR) cameras for seaport containers. The MEP collects and packages image data and associates it with the specific vehicle and manifest filing metadata obtained from other CBP IT systems.

CBP is committed to enhancing its mission effectiveness in securing the nation's borders, facilitating legitimate trade, and ensuring efficient travel. Non-Intrusive Inspection (NII) technologies are critical tools in achieving these objectives, providing essential capabilities for detecting illicit goods and threats. To that end, NII intends to modernize its infrastructure to enable CBP to revolutionize how NII data is utilized, including leveraging advanced technologies such as Artificial Intelligence and Machine Learning (AI/ML). All MEP system configurations and their subcomponents shall be capable of exporting all collected image and metadata in real-time to designated CBP IT systems and authorized external platforms. Real-time data export is essential to enable advanced artificial intelligence (AI) technologies to maximize scanner utility, support automated threat detection, and enhance operational efficiency.

The Functional Requirements Document (FRD) details MEP use cases and associated requirements for use of these systems by CBP officers inspecting cargo vehicles arriving at northern and southern land border POEs throughout the United States, as well as seaport POEs and United States Border Patrol (USBP) checkpoints. Different MEP configurations and use cases are documented within this document. Each POE will be evaluated on case-by-case basis and the design, infrastructure, and installation customized to accommodate the commercial NII technologies based on the POE

physical layout and existing infrastructure. See Attachment 03, *Design, Construction, and Installation Requirements*. The configurations and use cases in this document are generalized and may differ from the ones for a specific POE.

This MEP FRD is intended to be issued as Attachment 01 of a larger Statement of Work (SOW) and contracting package. Throughout this document are reference to other attachments to the overall SOW.

1.1 Mission Needs Statement (MNS) Gaps

The NII gaps from the NII Mission Needs Statement (MNS)¹ to be addressed by the integrated MEP System are listed in Table 2. Note that the “enhanced” CONOPS discussed in Gaps 4 and 7 are those associated with the NII-I Program and documented in the NII-I CORD.

Table 2 NII Mission Needs Statement Gaps to be addressed by MEP

#	Gap	Description	Type
1	NII Systems Integration	CBP NII imaging systems are currently not integrated with Radiation Detection Equipment (RDE), the CBP network, data center, local operational command centers nor CBP information technology enforcement applications and databases. Standalone NII large-scale systems do not support automated, efficient data exchange or real-time targeting, do not provide a consolidated threat picture for operators and do not support remote performance monitoring and system maintenance. By themselves, RPM's [Radiation Portal Monitors] are limited in detecting radiation from well-shielded weapons or radioactive material – such threats must also be scanned for anomalies by an NII imaging system to improve threat detection.	Materiel
4	Primary NII Image Scanning Capability / Capacity	Currently, NII image scanning is largely a secondary or targeted inspection process. Various operational vectors, including land border Privately Owned Vehicle (POV), truck cargo, seaports, mail, express consignment cargo, and air freight, do not perform NII scanning on a large percentage of incoming traffic / containers. The NII Program must identify and employ new, more efficient technology systems and associated CONOPs, which can offer an opportunity to greatly increase scan rates in these vectors without increasing wait times.	Materiel
7	Adaptation to New Technology and CONOPS	Newer Commercial NII technologies have the potential to improve capabilities and greatly increase NII scanning rates using enhanced CONOPS in pre-primary or primary inspection environments. POE facility space, layout and infrastructure is [sic] often not designed or built to	Materiel Facilities

#	Gap	Description	Type
		accommodate these new technologies or enhanced CONOPS.	
11	Aging and Obsolete NII Equipment	Many fielded systems, approximately 27 percent of the large-scale and 35 percent of small-scale NII systems, are in need of lifecycle replacement as they have exceeded the vendor defined service life or are already functionally obsolete.	Materiel
15	Chemical, Biological, and Explosive Detection Equipment	There is a lack of NII equipment capacity and capability to detect chemical, biological, and explosive threats and hazards. Additionally, the infrastructure and physical space at many POEs is insufficient to deploy and operate CBE detection systems.	Materiel Facilities

1.2 System Constraints

- Only active systems with generated X-ray sources will be considered.
- Per ANSI 43.17 2009, potential radiation dose to vehicle occupants that exceed 25 microRems (µR) per scan will not be accepted as stated in MEP FRD Section 11.2
- Any wireless communication, including Bluetooth that cannot be disabled by Administrator level operators, will not be accepted.
- MEP scanning systems exceeding threshold limits as stated in MEP FRD Section 6.1 will not be accepted.

2 MEP Configurations

The CBP POEs range from small (one to two lanes) to large (over 30 lanes), requiring a system capable of being configured to meet the needs of the various ports, including in pre-primary, post-primary, secondary, seaport or Border Patrol (BP) checkpoint inspection use cases. In some use cases, control stations are single operator workstations networked with a single MEP for a small-land POE port or seaport. In other cases, there are control stations with multiple portals multiplexed to multiple workstations for large ports used to facilitate a high MEP vehicle scanning throughput. MEP configurations include the following:

2.1 MEP Configuration 1 (No RFID or QR) for BP Checkpoint or Seaport

MEP Configuration 1 will be a BP checkpoint or seaport configuration integrated with the CBP network but does not include Decal and Transponder Online Procurement System (DTOPS) RFID or Quick Response (QR) code reader systems. At BP checkpoints LPR will be included for reading tractor/trailer license plates. For seaports,

OCR will be used to read container IDs. This configuration will employ a control booth adjacent to the MEP system, a site control center where MEPs at different locations are connected to accommodate multiple MEP operators and analyst workstations, or alternative setup. Configuration 1 will not include Under-Vehicle Back-Scatter System (UVBS). This configuration at minimum will be accompanied by the red light/green light function of the Traffic Control System (TCS). Radiation Portal Monitors (RPMs) are not integrated directly with the MEP, other than potentially a MEP blanking function, and RPM alarms are communicated to the primary officer as they are presently, via an annunciator within the booth. RPM data transmission and alarm adjudication are unchanged from the current process and therefore unaffected by the MEP system.

2.2 MEP Configuration 2 (with RFID and QR) for Land Border POE

MEP Configuration 2 is installed at a land border POE in pre-primary or post-primary location relative to existing primary inspection booths, RPMs, and integrated with the CBP network, cloud, law enforcement applications and a site control center. It will include TCS, DTOPS RFID, QR code, and LPR systems. Optional (future), the sensor technology should be open to new technologies identified for better data collection including Near Field Communication and Bluetooth Low Energy technology. Optional UVBS may be installed in this configuration. RPMs are not integrated directly with the MEP, other than potentially a MEP blanking function, and RPM alarms are communicated to the primary officer as they are presently, via an annunciator within the booth. RPM data transmission and alarm adjudication are unchanged from the current process.

2.3 MEP Configuration 3 (Virtual Primary) for Land Border POE

MEP Configuration 3 is the same as Configuration 2 except the “virtual” primary officer is located away from the traffic lane and communicates remotely with the driver via video conference. In this case, Configuration 3 will include the addition of “staging lanes” where vehicles must queue post-MEP inspection, await their Non-Intrusive Inspection (NII) data package adjudication result, and CBP officer direction. RPMs are integrated with the MEP system and RPM data is added to the MEP data package.

3 Requirements

3.1 Requirements Definitions

Threshold (T): The minimum level of operational performance that the Government is willing to accept is considered a threshold value.

Objective (O): An objective is considered a level of performance that significantly improves mission performance, safety, or supportability beyond that of the threshold value, and represents the maximum desired yield for system performance.

- Objective values are not required but are defined to provide guidance to the Contractor with respect to areas where increased capability is of interest to the Government.
- If objectives are defined, the interval between the objective and threshold values for a given parameter is the Government program manager's (PM) trade-space.
- If objective values are listed, the Contractor will evaluate them to determine if they can reasonably be achieved, whether there is a cost or schedule impact, and the extent of the impact. The impact of achieving objective values will be brought to the attention of the Government for review.

3.2 Requirements Numbering

Each requirement in this FRD is assigned a unique ID number of the form of XX.YY.ZZ.

- XX indicates the requirement category number.
- YY indicates the component number found in Table 3 or requirement group
- ZZ defines the individual requirement number.
- When there is no requirement for a given component in that requirement category, the ID is omitted entirely.

Two additional requirements groups are used in this document:

13. Security and Privacy
14. Reliability, Availability, and Maintainability (RAM).

For YY value, Section 13 uses a different set of requirement groups (not component number). These requirement groups are more appropriate to the Security and Privacy topic than component numbers.

Section 14 has no specific numbered requirements. Instead it describes design considerations necessary to meet Lifecycle Sustainment and Support Requirements attachment of the SOW.

4 MEP Use Cases

The use case scenarios are presented as a general reference point for the following operations:

1. Border Patrol Checkpoint
2. Seaport Inspection
3. Land Border POE with Pre-Primary or Post-Primary Scan

The specifics of the scenarios will depend on each location's unique layout/constraints and modified accordingly. The process steps for "X-ray opt-out" and control center analyst functions are included within the inspection use cases.

4.1 MEP Use Case 1 – Border Patrol Checkpoint

The use case provides a general description of CBP operations when scanning cargo vehicles at a BP checkpoint. This use case begins when a conveyance is directed to and/or approaches the MEP unit and enters the traffic control system at its first control point. This use case ends when the conveyance is cleared for checkpoint exit or an anomaly is detected and the conveyance is referred for further inspection

Use Case Pre-Condition:

- 1) The MEP System is ready for operation and MEP operator/analyst(s) are present in either an adjacent control booth or site control center.
- 2) The vehicle/conveyance is directed to the MEP by a BP agent based on a probable cause determination or by vehicle driver voluntary agreement when asked. Note that this pre-condition is different from land border POEs where CBP has different authorities and the goal is 100% NII scanning.

Use Case Post Condition:

- 1) The conveyance is released and cleared for checkpoint exit, and the results of the cargo processing are documented in the MEP System and the required Government systems, or
- 2) BP detains shipment after detecting an anomaly and is referred for further investigation.

Use Case Steps

- 1) The truck enters the BP checkpoint and is initially screened by BP agent/canine. If probable cause exists or if the BP agent asks the driver to allow the vehicle to be scanned (i.e., X-ray) and the driver agrees, the vehicle is designated for MEP scan.
- 2) The truck approaches the MEP System, is screened for oversize and diverted as required.
- 3) The truck otherwise continues to approach the MEP System and vehicle presence is detected at the first traffic control point and enters into positive control by the Traffic Control System (TCS). Positive control of the traffic occurs at the point where conveyances proceed through the MEP system and are directed by lights, signage, control arms, and other measures.
- 4) The TCS sensors confirm that a 'cleared to proceed' condition exists by confirming an open traffic lane through the MEP Unit and clearance of the required post-scan area such that the conveyance fully transits the MEP in one continuous movement.
- 5) Upon receiving a 'green light' the truck begins transit through the MEP Unit.

- 6) The MEP System TCS light behind the active truck in the MEP turns red to stop the next vehicle at the first traffic control point.
- 7) The truck continues to move through MEP Unit at the designated scanning speed.
- 8) The truck speed is detected, and visual feedback of speed is provided to the driver.
- 9) When the MEP Unit detects a truck entering the scanning system:
 - a. The MEP Unit reads the front tractor license plate and the trailer license plate.
 - b. MEP Unit starts driver-safe transmission X-ray scanning of cab with low energy / flux, or reflective (back-scatter) X-ray scanning
 - c. When MEP Unit detects the end of cab, MEP switches to X-ray scans at high flux using alternating high / low energies.
 - d. If “end of cab” is not detected the entire truck is scanned at driver-safe low energy / flux and a warning indicator is supplied to the MEP operator and analyst in the control booth /NII control center.
 - e. If MEP detects a second trailer, MEP X-ray scans the second trailer at high flux using alternating high / low energies.
 - f. If the cab of the truck stops within the MEP or advances at a speed that could potentially expose vehicle occupants to radiation dose above 25 µR per scan, MEP system stops scanning.
- 10)When MEP Unit detects that the truck has exited the scanning area:
 - a. MEP Unit stops X-ray scanning.
 - b. MEP Unit stops capturing undercarriage imagery / X-ray.
- 11)When the truck clears one complete truck / trailer length (approximately 100' minimum) from the MEP, the MEP system TCS indicator reverts to green, signaling the next truck to advance into the MEP. It allows for the length of truck and trailer, including the potential for multiple trailers towed by a single tractor. The MEP will be configurable to accommodate multiple trailers at sites which have the physical space to accommodate such loads.
- 12)The truck proceeds to secondary staging area to await adjudication of images. Traffic controls or agents direct the truck's movement and where to stop.
- 13)The MEP System creates and timestamps the data package and assigns a unique case record number. The MEP data packages plus other MEP health and performance data are transmitted to the CBP network and data center/cloud.
- 14)The MEP data package is 'posted' to the MEP queue in the control booth / NII control center.

- 15) A control booth / NII control center analyst is automatically assigned a MEP scan image in the order scanned, which triggers the analysis process.
- 16) The control booth / NII control center analyst provides a 'red' suspect referral or 'green' non-suspect release recommendation and annotates the preliminary decision in the Government system. The MEP operator/analyst annotates the image with overlay graphics to indicate the location of the anomaly if present.
- 17) If 'red,' the MEP scan image is available to agents for reference to perform follow on enforcement actions.
- 18) The truck driver is informed of the next step in the process by face-to-face interaction with a roving agent.
- 19) The agents complete the process and enter the final results into the Government system.
- 20) This use case ends when the truck driver either leaves the checkpoint area and proceeds to exit as a cleared load or is detained for further inspection.

4.2 MEP Use Case 2 - Seaport Inspection

Scanning with booth / officer and IT integration but without RPM integration. Seaport may have RPM considerations needed and interference discussions (as stated in Configuration 1).

Use Case Steps

- 1) The truck with targeted sea container approaches the MEP System, is screened for oversize and diverted as required.
- 2) The truck advances and vehicle presence is detected at the first traffic control point as the truck enters into positive control by the TCS. Positive control of the traffic occurs at the point where conveyances proceed through the MEP system and are directed by lights, signage, control arms, and other measures.
- 3) The TCS sensors confirm that a 'cleared to proceed' condition exists by confirming an open traffic lane through the MEP Unit and clearance of the required post-scan area such that the conveyance fully transits the MEP in one continuous movement.
- 4) Upon receiving a 'green light' the truck begins transit through the MEP Unit.
- 5) The MEP System TCS light behind the active truck in the MEP turns red to stop the next shipment at the first traffic control point.
- 6) The truck continues to move through MEP Unit at the designated scanning speed.
- 7) The truck speed is detected, and visual feedback of speed is provided to the driver.
- 8) When the MEP Unit detects a truck entering the scanning system:
 - a. The sea container ID number on sides or back is read using the Optical Character Recognition (OCR) system.

- b. After truck cab passes and when MEP Unit detects the start of the trailered sea container, MEP X-ray scans the container at high flux using alternating high / low energies.

Note: Scan of the cab is not performed in this use case. Seaport MEP Configuration 1 does not include RFID, LPR, QR, or UVBS equipment.

- c. If the cab of the truck stops within the MEP or advances at a speed that could potentially expose vehicle occupants to radiation dose above 25 μ R per scan, MEP system stops scanning.
- 9) When MEP Unit detects that the truck has exited the scanning area, the MEP Unit stops X-ray scanning.
 - 10) When the truck clears one complete truck / trailer length (approximately 100') from the MEP, the MEP system TCS indicator reverts to green signaling the next truck to advance into the MEP.
 - 11) The truck proceeds to a staging area to await image adjudication.
 - 12) The MEP system creates and timestamps the data package and assigns a unique case number. The MEP data package plus other MEP health and performance data transmitted to the CBP network and data center/cloud.
 - 13) The MEP scan image is 'posted' to the MEP analysis queue in the control booth / NII control center.
 - 14) A control booth / NII control center analyst is automatically assigned a MEP scan image in the order scanned, which triggers the analysis process.
 - 15) The control booth / NII control center analyst provides a 'red' suspect referral or 'green' non-suspect release recommendation and annotates the preliminary decision in the Government system. The MEP operator/analyst annotates the image with overlay graphics to indicate the location of the anomaly.
 - 16) The officer in the staging area is informed of the analysis decision
 - 17) If 'green,' the hold for NII inspection is cleared. The officer in the staging area directs the truck driver proceeds to the exit gate or returns the container to the storage yard.
 - 18) If 'red,' the officer may do a quick inspection (e.g., tailgate) to clear the truck. If cleared, the 'red' referral is changed to 'green' and the officer in the staging area directs the truck driver proceeds to the exit gate or returns the container to the storage yard.
 - 19) If the referral is not resolved, the truck is directed to the seaport Centralized Examination Station (CES) or other location for Secondary inspection.
 - 20) This use case ends when the truck driver exits the examination area and proceeds to the exit gate as a cleared load, or the container is referred for Secondary inspection.

4.3 MEP Use Case 3 – Land Border POE with Pre-Primary or Post-Primary Scan

Land border pre-primary or post-primary scanning with primary booth / officer and IT integration but without RPM integration. The use case provides a general description of CBP POE operations when scanning cargo at a land border crossing. This use case begins when a conveyance approaches the MEP Unit and enters the traffic control system at its first control point. Positive control of the traffic occurs at the point where conveyances proceed through the MEP system and are directed by lights, signage, control arms, and other measures. This use case ends when the conveyance is cleared for POE exit, or an anomaly is detected, and the conveyance is referred for further inspection.

Use Case Pre-Condition:

- 1) The MEP System is ready for operation and MEP operator / analyst(s) are present in a site control center. The primary booths are manned by primary officers.
- 2) The vehicle / conveyance is directed to the check-in kiosk and MEP by the traffic control system in pre-primary or post-primary.

Use Case Post Condition:

- 1) The conveyance is released and cleared for POE exit, and the results of the cargo processing are documented in the MEP system and the required Government systems, or
- 2) The conveyance is referred for Secondary inspection.

Use Case Steps:

- 1) Truck enters the POE, approaches the MEP System, is screened for oversize and diverted as required.
- 2) The truck otherwise advances to a “check-in” kiosk located in each entry lane before the MEP system outfitted with QR code, RFID and license plate read capability, to initiate the creation of a MEP data package and comes to a stop. (If no opt-out go to Step 5).
- 3) Alternately, the driver can choose to opt-out of MEP scanning. If so:
 - a. Before the MEP check-in kiosk, driver follows traffic control directional signage to maneuver the truck to an opt-out lane leading to a primary booth, passes through RPM and approaches primary officer. Any RPM alarms are sent directly to existing primary booth annunciator in the opt-out lane.
 - b. Alternately, if no opt-out lane is available, and wishes to opt-out of X-ray scan, driver communicates intentions to control center officer via

kiosk audio/visual connection and MEP X-ray function is disabled. Balance of MEP scanning function still occurs.

- c. Truck can be directed by primary officer to:
 - i another secondary non-drive-through X-ray system, or
 - ii the secondary warehouse for manual processing, or
 - iii POE Exit.
- 4) Truck presence is detected at kiosk; CBP GEN 2 DTOPS RFID tag is read if available, and all front and rear license plates are automatically read. Driver presents QR code (if available) on either paper or smart phone to QR code reader.
RFID tag ID and user memory; QR code; and license plates are recorded.
- 5) If driver does not opt-out, the truck approaches the MEP system and vehicle presence is detected at the first traffic control point. If a previous truck is still transiting the MEP system, the approaching driver receives a red light from the TCS and stops.
- 6) The truck enters into positive control by the TCS.
- 7) The TCS sensors confirm that a 'cleared to proceed' condition exists by confirming an open traffic lane through the MEP Unit and clearance of the required post-scan area such that the conveyance fully transits the MEP in one continuous movement.
- 8) Upon receiving a 'green light', the truck begins transit through the MEP Unit.
- 9) The MEP System TCS light behind the active truck in the MEP turns red to stop the next conveyance at the first traffic control point.
- 10) The truck continues to move through MEP Unit at the designated scanning speed.
- 11) The truck speed is detected, and visual feedback of speed is provided to the driver.
- 12) When the MEP Unit detects a truck entering the scanning system:
 - a. The MEP Unit reads the front and rear license plates.
 - b. The MEP Unit reads the CBP DTOPS RFID tag EPC (memory bank #1) and user memory (memory bank #3) coded with ACE trip, date of arrival and license plate information, if equipped with an RFID reader.
 - c. A blanking signal is sent to the RPM(s) (if so equipped/configured).
 - d. The MEP Unit starts transmission X-ray scanning of cab with low energy / flux or reflective (back-scatter) X-ray scanning if so equipped.

- e. If equipped, the MEP Unit UVBS starts capturing undercarriage back-scatter imagery.
 - f. When MEP Unit detects the start of trailer, MEP X-ray scans at high flux using alternating high / low energies.
 - g. If “start of trailer” is not detected, the entire truck is scanned at low energy / flux and a warning indicator is supplied to the MEP operator and analyst in the control center.
 - h. If MEP detects a second trailer, MEP X-ray scans the second trailer at high flux using alternating high / low energies.
 - i. If the cab of the truck stops within the MEP or advances at a speed that could potentially expose vehicle occupants to radiation dose above 25 µR per scan, MEP system stops scanning.
- 13) When MEP Unit detects that the truck has exited the scanning area:
- a MEP Unit stops X-ray scanning.
 - b MEP Unit stops capturing undercarriage UVBS imagery.
- 14) When the truck clears one complete truck / trailer length (approximately 100') from the MEP, the MEP system TCS indicator reverts to green signaling the next truck to advance into the MEP. The MEP will be configurable to accommodate multiple trailers at sites which have the physical space to accommodate such loads.
- 15) The truck exits from positive traffic control upon fully exiting the MEP Unit.
- 16) The truck proceeds to next available primary booth, passing through an RPM for a given primary booth/lane. Any RPM alarms are sent directly to existing primary booth annunciator.

Note: The exact order of events (RPM scan vs MEP scan) and relative positioning of the RPM in relation to the MEP will be determined specifically for each site installation, with consideration given to eliminating RPM interference caused by the MEP X-ray.

Note: The process steps listed above are similar for post-primary MEP installation with the truck first encountering an officer in a primary booth and then proceeding to the post-primary MEP system. Following post-primary MEP scan, the truck proceeds to a staging lane where it awaits scan image analysis, officer adjudication and release / refer decision. The staging lane will have a kiosk supporting LPR, RFID and QR readers as well as audio-visual communications capability to identify the truck for the CBP control center analysts and allow communications with the driver.

- 17) The MEP System creates and timestamps the data package and assigns a unique case record number. The MEP data packages plus other MEP health and performance data are transmitted to the CBP network and data center/cloud.
- 18) The MEP data package is 'posted' to the MEP image queue in the site NII control center.
- 19) The NII control center analyst provides a 'red' suspect referral or 'green' non-suspect release recommendation and annotates the preliminary decision in the Government system. The MEP operator/analyst annotates the image with overlay graphics to indicate the location of the anomaly.
- 20) The NII control center analyst decision is communicated to the primary booth officer by the Government system.
- 21) When the truck approaches a primary booth, the primary booth officer enters tractor LP into the Government system, pulls up the ACE trip shipment record and reviews the MEP image analysis decision. Alternately, if available, the primary booth RFID / QR reader system will read the truck DTOPS RFID tag or QR code and automatically call up the ACE trip shipment record and the MEP image analysis adjudication. If the MEP scan image is not yet fully adjudicated by the control center officer, the primary officer receives a "pending" indication. The primary officer can await an analyst adjudication decision or send the "pending" truck to secondary.
- 22) The primary booth officer otherwise performs the normal primary inspection and makes the final decision on whether to release the shipment or refer for secondary inspection.
 - a. 'Green' non-suspect MEP data adjudications may be referred to secondary for other reasons at the primary booth officer's discretion.
 - b. 'Red' suspect MEP data adjudication decisions must be sent to secondary.
- 23) Note: For post-primary MEP installation, the final release / refer decision is made by the control center analyst in coordination with the primary officer's input. If the conveyance is not referred to secondary, the truck is released.
- 24) This use case ends when the truck driver exits the POE or is referred for secondary inspection.

5 Functional Requirements

Table 3 lists the major components of the MEP system. In addition to these major components the system includes associated ancillary hardware and software elements.

Table 3 MEP System Functional Components

Component	Functional Component Description
MEP System	The MEP System refers to the entire set of MEP Units, systems, and subsystems installed at a Land Border POE, Seaport, or BP Checkpoint

MEP Unit [Single Lane]	The MEP Unit refers to the systems and subsystems, which are installed in one traffic lane to execute the full scanning operation. The MEP system is a drive-through, multi-energy X-ray system. The MEP system will include integrated RFID transponder reader and license plate reader (LPR) for use at land border POE, or container number optical character recognition (OCR) system for use at seaport POE, to uniquely identify the MEP data package for the specific vehicle/container being scanned. The land POE drive-through systems may include an Under Vehicle Back-Scatter system (UVBS).
MEP Unit X-Ray System	X-ray scanning system capable of providing images and material characterization of the tractor, trailer, container and cargo. As an option, the MEP Unit X-ray may include a backscatter X-ray system (in addition to primary transmission X-ray system).
Under-Vehicle Back-Scatter System (UVBS) - Objective – Optional Contract Provision	Backscatter X-ray scanning system capable of providing images of the undercarriage of vehicles.
License Plate Reader (LPR) System (Configuration 2 & 3, Land Border)	Identification system that uses OCR technology to read the truck/tractor (front) and rear (trailer) license plates. Provides jurisdiction (e.g., state, province) and plate alphanumeric ID.
RFID Reader System (Land Border configuration)	Identification system that uses RF signals to read passive (backscatter) transponders (tags) mounted on the truck/tractor. CBP issues DTOPS RFID transponders to enrolled vehicles in Free and Secure Trade (FAST), a trusted partner program that provides expedited screening and clearance. This system reads the RFID transponder on the vehicle and provides the associated identification number to the MEP Unit.
Optical Character Recognition (OCR) System	Identification system that uses OCR technology to read the container ID on the sides and back of multimodal shipping containers at seaports.
Check-In / Staging Lane Kiosk	A kiosk with integrated RFID, QR code and license plate reader capability to identify specific trucks and shipments at POE check-in or at a post-MEP staging lane. Kiosk may include two-way audio and video feed to allow control center interaction with the driver.
Traffic Control System (TCS)	The Traffic Control System / subsystems (TCS) will ensure the efficient and effective flow of traffic without negatively impacting the overall flow of cargo through the POE. This capability may include lights, signage, control arms, pavement striping, lane barriers, and other elements as required. The TCS will be integrated with MEP operation and the MEP LAN system. It will include a traffic/lane controller to automatically and actively guide traffic through the system steps and manage lane closures, MEP system outages, etc., with operator or smart queuing system input.
Vehicle Sensors	Sensors used to detect the presence of vehicles in a specific position in a lane. Examples include inground inductive loops or break-beam sensors.

CCTV and Camera Systems	Closed-circuit TV systems used for surveillance. Has the capability to record and view video. Uses point-tilt-zoom (PTZ) video cameras.
Data Storage (DS)	Local computer data storage to preserve images and data packages.
MEP Work Center	Workstations where CBP officers and agents can view and enhance (e.g., zoom, contrast) X-ray images to look for the presence of anomalies that would warrant further inspection of vehicles or cargo.
NII Control Center	The NII Control Center will allow for remote operational control of the MEP System. Remote control includes the operation of the MEP Unit(s), the traffic control system, required communications to support the workflow, situational awareness (CCTV system displays), and any other requirements to fully operate the MEP System and perform analysis of MEP data. In some secondary locations, the control center can be a simple operator / analyst booth located adjacent to the MEP system. In other applications, it can be a stand-alone facility or dedicated office space within an existing POE or regional facility. Within the control center, there can be one or more work centers, consisting of MEP operator / analysis workstations and an associated CBP networked workstation hosting law enforcement applications such as ACE-TMM, TECS, etc.

5.1 MEP Unit Functional Requirements

The MEP Unit refers to the systems and subsystems which are installed in one traffic lane or one inspection/scanning area to make up the full scanning operation.

- 5.1.1 The MEP Unit shall be designed and configured for operation from a CBP booth or command center.
- 5.1.2 The MEP Unit, including all associated equipment, shall accommodate vehicles up to 15' 0" in height (T).
- 5.1.3 The MEP Unit, including all associated equipment, should accommodate vehicles up to 16' 0" in height (O).
- 5.1.4 The MEP Unit, including all associated equipment, shall have a tunnel width that is at least 12' 0" wide (T).
- 5.1.5 The MEP Unit, including all associated equipment, shall have a tunnel width that is at least 13' 6" wide (O).
- 5.1.6 The MEP Unit shall scan multiple trailers towed by a single tractor when site conditions permit this operation.

5.2 MEP Unit X-Ray System Functional Requirements

- 5.2.1 The MEP Unit shall use active X-Ray technology.
- 5.2.2 The MEP Unit shall perform X-ray transmission scan of the trailer or trailers at high flux using alternating high/low energies and cab at low flux/energy as described in Sections 7.3 and 11.2.

- 5.2.3 The MEP Unit shall display the entire length of the vehicle (front to end of last trailer), from the ground to the roof for the side views and the complete length and width of the vehicle for the top-down view or bottom-up view.
- 5.2.4 The MEP Unit should have a backscatter X-ray scan capability from top down and for each side of the scanned vehicle to supplement the high energy transmission scan capability (O).
- 5.2.5 The MEP Unit should support alternative beam geometries, other than perpendicular to the trailer/container, to enhance detection of contraband within fore and aft walls, doors, etc. (O).
- 5.2.6 The MEP Unit shall be capable of material discrimination using the multi-energy capability to distinguish, at a minimum, between organic (low-z), mix of inorganic/composite (medium-z) and inorganic (high-z) materials.
- 5.2.7 The MEP Unit shall allow the operator-analyst to toggle material discrimination colors on and off.
- 5.2.8 The MEP Unit shall distinguish organic ($\text{low } Z \leq 10$), intermediate ($10 < Z < 20$), and metallic ($Z \geq 20$) materials using different colors for their display. Material categories are to be color-coded as follows: organic (orange), intermediate (green) and metallic (blue) (T). Shades of these colors are to be used to indicate thickness. All colors should be in compliance with the Unified File Format, Version 3. Material Discrimination testing will be conducted per Attachment 5, Section 1.6.
- 5.2.9 The MEP Unit should distinguish Heavy Metals ($Z \geq 70$) (Color not defined by CBP) (O).

5.3 Under-Vehicle Back-Scatter System (UVBS) - Objective – Optional Contract Provision Functional Requirements

- 5.3.1 If included, the UVBS reflective X-ray shall be integrated into the scanning operation of the MEP Unit to perform high resolution scan of the undercarriage of a standard width tractor-trailer configuration and variable under vehicle heights.
- 5.3.2 If included, the UVBS shall capture the complete length and width of the undercarriage of the vehicle.
- 5.3.3 If included, the UVBS scan image shall be spatially aligned with the MEP x-ray (transmission/back-scatter) scan images.
- 5.3.4 If included, the under-vehicle inspection system shall incorporate an automated mechanism to remain consistently free of obstructions, such as dust and debris, ensuring the reliable acquisition of X-ray images during each inspection.

5.4 License Plate Reader (LPR) System Functional Requirements (Land Border configuration)

- 5.4.1 The LPR system shall be capable of reading all front license plate of the tractor and all rear license plate of the last trailer. Where license plate comply with

American Association of Motor Vehicle Administrators (AAMVA) License Plate Standard, Edition 3 (or latest standard).

- 5.4.2 The LPR system shall provide images of the front and rear areas of the vehicle, each of which includes the complete plate to allow visual confirmation of the plate number and jurisdiction.

5.5 RFID Reader System Functional Requirements (Land Border configuration)

- 5.5.1 The RFID system shall be capable of reading GEN 2 RFID tags, per ISO-18000-63 version 2 specification, Information Technology - Radio Frequency Identification for Item Management - Part 63 with EPC Global / GS 1 GDTI-96 tag format.
- 5.5.2 The RFID system shall filter out non-DTOPS tags based on CBP defined header in memory bank #1 EPC Global / DTOPS registration number.

5.6 Optical Character Recognition (OCR) System Functional Requirements

- 5.6.1 The OCR shall be provided for seaport configurations in full, integrated operability with the MEP Units.
- 5.6.2 The OCR shall read the Container ID used to identify each container passing through the MEP Unit.
- 5.6.3 The OCR system shall provide images of both sides and rear of the container that each includes the complete ID to allow visual confirmation of the optical character read.

5.7 Check-In / Staging Lane Kiosk Functional Requirements

- 5.7.1 The MEP System shall include provision to read RFID, QR code and license plates at the Check-In and Staging Lane Kiosks.
- 5.7.2 The MEP System shall read all of front tractor license plates, all plates of the last trailer, RFID and driver-presented QR codes at the Check-In and Staging Lane Kiosks.
- 5.7.3 The QR code reader shall be able to read driver-presented QR codes at the MEP entry tunnel.
- 5.7.4 The staging lane kiosk shall include two-way High-Definition (HD) audio with the control center officer to allow direct communication with the truck driver to ask questions or provide instruction in compliance with Centralized Area Video Surveillance System (CAVSS) requirements.
- 5.7.5 The staging lane kiosk shall include one-way HD video connection with the control center officer to allow observation of the truck driver in compliance with CAVSS requirements.

5.8 Traffic Control System (TCS) Functional Requirements

- 5.8.1 The traffic control systems/subsystems (TCS) shall ensure the efficient and effective flow of traffic without negatively impacting the overall flow of vehicles and cargo through the POE.
- 5.8.2 The TCS shall include positive control of each conveyance before entering the MEP lane. Positive control of the traffic occurs at the point where conveyances proceed through the MEP system and are directed by lights, signage, control arms, and other measures.
- 5.8.3 The TCS shall control the conveyance through the MEP lane.
- 5.8.4 The TCS shall control the area after the scan to allow the conveyance of the proper stopping distance including the post-MEP staging lane area, as applicable.
- 5.8.5 If required due to traffic congestion, the TCS shall allow the truck to safely come to a stop after fully exiting the MEP Unit.
- 5.8.6 The TCS shall not require the truck to stop after the scanning sequence is started.
- 5.8.7 The TCS shall be capable of directing traffic for driver opt-outs and oversize vehicles.
- 5.8.8 The TCS shall control traffic flow when exception conditions occur. Note: Examples of exception conditions are: Lane Closure (e.g. disabled truck); Response to emergency conditions either by CBP Officers or by other emergency services.
- 5.8.9 The MEP TCS shall be configured to allow vehicles to pass through the MEP Unit if the MEP Unit is disabled.
- 5.8.10 The TCS shall be configured to include pre-set, automatic traffic pattern sequences to accommodate standard and non-standard traffic patterns and exception conditions (e.g. MEP Unit failure, etc.) initiated by CBP officers in the NII Control Center.
- 5.8.11 The TCS shall be designed and configured for unmanned operations during standard operating scenarios.
- 5.8.12 The TCS shall be designed and configured for unmanned operations for identifiable non-standard operating conditions such as a MEP Unit not being available for scanning.
- 5.8.13 The TCS shall be configured to maximize throughput for all operational MEP lanes and provide CBP Officers with the ability to actively route traffic past the MEPs when necessary to maintain optimal primary operations.
- 5.8.14 The TCS shall take inputs from lane capacity at the primary booth and other sensors to automatically assign trucks to MEP lanes, bypass lanes, or specific booths to maximize throughput and maintain optimal primary operations. (O)
- 5.8.15 The TCS shall optimize traffic flow for all operational MEP lanes. (O)

- 5.8.16 The TCS shall include logic-based routing to optimize traffic flow. (O)
- 5.8.17 CBP Officers shall have the ability to route traffic past MEP lanes to maintain optimal primary operations.

5.9 Vehicle Sensors Functional Requirements

- 5.9.1 The MEP Unit(s) shall include vehicle presence sensors or other means that monitor and record the progress of the vehicle as it approaches, passes through, and exits each MEP Unit and indicates whether a lane is clear or occupied.
- 5.9.2 The MEP Unit(s) shall measure the vehicle's instantaneous speed to the nearest 0.1 MPH from 0 to 15 MPH as it approaches and passes through each MEP Unit.

5.10 Closed Circuit Television (CCTV) and Camera Systems Functional Requirements

- 5.10.1 The CCTV and camera systems shall provide a Fixed CCTV system for each MEP Unit.
- 5.10.2 The Fixed CCTV system shall capture live video of vehicles traveling through the MEP Units.
- 5.10.3 The Fixed CCTV system shall provide full 360-degree coverage of the MEP Unit operating area.
- 5.10.4 The CCTV and camera systems shall provide Pan-Tilt-Zoom (PTZ) cameras for each MEP Unit.
- 5.10.5 PTZ cameras positioned at the entrance and exit of each MEP Unit shall provide sufficient resolution and image quality to permit qualitative descriptions (cab color, make, model; container color and length) of all containers and vehicles that pass through the MEP.
- Note: The CBP Border Security Deployment Program (BSDP) has established an approved specification for surveillance video cameras and recording called CAVSS. Any camera covering interactions between CBP officers and drivers must comply with CAVSS requirements.
- 5.10.6 The CCTV and camera systems shall provide elevated surveillance PTZ cameras for wide-area situational awareness
- 5.10.7 The surveillance PTZ cameras shall have the capability to focus using pre-programmed presets.
- 5.10.8 All camera views (CCTV, MEP PTZ, surveillance PTZ) shall be automatically recorded.
- 5.10.9 The CCTV and camera systems shall be capable of providing screenshots at any time.

5.11 Data Storage (DS) Functional Requirements

- 5.11.1 The MEP System Data Storage (DS) shall store the data packages (including X-ray images, IDs, and metadata) sent to the NII Control Center or Cloud.
- 5.11.2 The MEP System DS shall store X-Ray images in an uncompressed data format.
- 5.11.3 The MEP System DS shall retain raw X-Ray image data for high and low energies for each data package in addition to any proprietary/processed X-Ray image format.
 - Note: “Raw X-Ray image data” means the original x-ray image in its native format without any type of processing or enhancement.
- 5.11.4 The MEP System DS stored images shall retain the full dynamic gray-level range of the X-Ray image
- 5.11.5 The MEP System DS shall store data packages for a rolling first-in first-out period of thirty (30) days.
- 5.11.6 The MEP System DS shall support operator-defined queries, including standard reports for scans performed, scans per day/per analyst and other Government defined metrics.
- 5.11.7 The MEP System DS shall provide an option at the Administrator level for auto deleting files that are over 30 days old.
- 5.11.8 The MEP System DS shall have a full capacity backup system that is configured to automatically backup files at designated intervals.
- 5.11.9 The MEP System DS shall delete the oldest data packages to restore room for new packages when storage reaches 99% of its maximum storage capacity.
- 5.11.10 The MEP System DS shall not over-write any previous decisions or annotations made by any analyst due to any analyst manipulations or actions at any System workstations, regardless of location.

5.12 MEP Work Center Functional Requirements

- 5.12.1 The MEP Work Center(s) shall allow all the required functions to fully adjudicate the X-ray images.
- 5.12.2 The MEP Work Center(s) shall provide the ability to do the following:

- enhance the image(s),
- change the magnification,
- select different color palettes,
- optimize the image's contrast and brightness,
- sharpen the image,
- highlight potential areas of interest,
- annotate the file for secondary inspection,
- input decisions on shipment adjudication, and
- perform other standard and required operations to meet the objective of detecting contraband and anomalies. (T)

5.12.3 The MEP Work Center(s) shall display complete images free of vertical or horizontal lines, any noise and any data or corruption that may confuse or hinder the operators' display, i.e., missing or incomplete data such as partial vehicle scans.

5.12.4 The System shall display the username and time with the image and the data package.

5.12.5 The System shall tag each scan analysis with the username who conducted it.

5.13 NII Control Center Functional Requirements for Local Network Deployments

5.13.1 The NII Control Center Situational Wall (SW) shall include two (2) scan queues on separate monitors as described below:

- Primary scan queue shall be displayed on a screen for the un-adjudicated scans from the MEP Units.
- Secondary scan queue shall be displayed on a screen for all shipments which are 'red lighted' and sent to the secondary warehouse for secondary examination.

5.13.2 Supervisors in the NII Control Center with Administrator accounts shall control the MEP System.

- Note: Control of the system includes the operation of the MEP Unit, NII Control Center SW, the traffic control system, any required communications to support the workflow, situational awareness (CCTV system displays), and any other requirements to fully operate the MEP System.

5.13.3 For each MEP Unit, the NII Control Center SW shall allow display for each CCTV/PTZ system so that the images associated with each MEP unit (i.e. lane) are easily identified.

6 Physical Requirements

6.1 MEP System Physical Requirements

- 6.1.1 The MEP System shall include computers and server(s) with enough memory, speed and other features to meet all the requirements of this FRD and supporting contract SOW, its attachments, and any related references.
- 6.1.2 Printers shall be capable of high-definition color printing.

6.2 MEP Unit Physical Requirements

- 6.2.1 The MEP Unit shall operate whether installed in a pre-primary, post-primary or secondary position.
- 6.2.2 The MEP Unit shall be able to be installed within a defined footprint no wider than 29 feet, no longer than 35 feet and no higher than 21.5 feet (T). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.
- 6.2.3 The MEP Unit should be able to be installed within a defined footprint no higher than 20 feet (O). These dimensions do not include any fencing or shielding walls and other auxiliary equipment or exit distance.

6.3 Under-Vehicle Back-Scatter System (UVBS) – Objective Physical Requirements

- 6.3.1 If included, the UVBS shall be flush mounted with the road.
- 6.3.2 If included, the UVBS physical dimensions (length, width, depth) should fit within the lane or underneath the secondary lot canopy.

6.4 Check-In / Staging Lane Kiosk Physical Requirements

- 6.4.1 The Check-In Kiosk shall be located at the MEP tunnel where the truck comes under positive control of the Traffic Control System (TCS). Positive control of the traffic occurs at the point where conveyances proceed through the MEP system and are directed by lights, signage, control arms, and other measures.
- 6.4.2 The Staging Lane Kiosk shall be located after the MEP where trucks queue awaiting control center adjudication of their scans and further instruction.

6.5 Traffic Control System (TCS) Physical Requirements

- 6.5.1 The TCS shall include all lanes leading into or bypassing the MEP Units and to the agreed upon distances before and after the MEP Units.
- 6.5.2 As applicable, the TCS controller shall be centrally located in the NII Control Center with a separate operating workstation.
- 6.5.3 The TCS shall be located and configured to allow a truck to achieve required minimal scan speed prior to entering the MEP Unit scan tunnel.

- 6.5.4 The TCS should achieve the traffic control goals without the use of gate arms (O).
- 6.5.5 The TCS shall have mechanical barriers to restrict oversized conveyances defined by height (5.1.2, 5.1.3) and/or width (5.1.4, 5.1.5) prior to entering the MEP lane to prevent collision damage to the MEP Unit.
- 6.5.6 The TCS shall include active signage to direct vehicles to open MEP lanes.
- 6.5.7 The TCS shall include active signage to direct vehicles past MEP lanes when necessary to maintain optimal primary operations.
- 6.5.8 The TCS shall include active signage to direct vehicles to open primary lanes after the MEP lanes.
- 6.5.9 The TCS shall include a minimum of one bypass lane for every two MEP lanes to ensure sufficient bypass capacity.
- 6.5.10 The TCS shall include license plate readers and manual QR readers.
- 6.5.11 The TCS shall include infrastructure (2" / 4" power/comms conduit, and 48 strand fiber) to support additional technologies, such as facial recognition cameras, touch screen monitors, and other future systems.
- 6.5.12 The TCS shall deploy the infrastructure for these additional technologies prior to the first set of traffic control structures and for all bypass lanes.

6.6 CCTV and Camera Systems Physical Requirements

- 6.6.1 The Fixed CCTV system shall be a high-definition Closed-Circuit Television System (CCTV) for each of the MEP Units, and associated operations, that is compliant with CBP CAVSS and approved by CBP BSDP.
- 6.6.2 The MEP Unit PTZ cameras shall be placed to allow for viewing of the license plates of the tractor and the trailer and provide enough range to view the tractor cab and the rear of the trailer.
- 6.6.3 The MEP Unit PTZ cameras shall include cameras at the entrance and exit of each MEP Unit
- 6.6.4 The Elevated PTZ cameras shall be attached to one or more MEP Units to provide for full situational awareness of the traffic lanes leading up to the MEP Units and the lane areas between the MEP Units and the Primary Booths.
- 6.6.5 The MEP System shall include a high-resolution color monitor (minimum 32").
- 6.6.6 The MEP Network Video Recorder (NVR) with a minimum 30-day recording ability capable of playing back any camera view.

6.7 MEP Work Center Physical Requirements

- 6.7.1 The MEP workstations shall comply with DHS Instruction 119-02-03, "DHS Workspace Standard"

6.8 NII Control Center Physical Requirements

- 6.8.1 The NII Control Center shall include an administrative workspace, where supervisors with administrator accounts can manage workload situations, emergencies, respond to system alarms (e.g. gate-arm crash), activate and deactivate MEP lanes, effect an orderly shutdown in case of power failures, etc.
- 6.8.2 The NII Control Center shall comply with DHS Instruction 119-02-03, "DHS Workspace Standard"; and 2023 CBP LPOE Design Standards and CBP security requirements and CBP OIT requirements.
- 6.8.3 The NII Control Center SW shall include one (1) high-definition screen TV for monitoring news and weather channels.

7 Performance Requirements

General performance relates to quantity, speed, and other general performance characteristics of the system.

7.1 MEP System Performance Requirements

- 7.1.1 The MEP System data shall be available to the MEP Work Center in five (5) seconds (T) after the truck scan is complete.
- 7.1.2 The MEP System data should be available to the MEP Work Center in three (3) seconds after the truck scan is complete (O).
- 7.1.3 The MEP System data shall begin to upload to the NII Cloud in two (2) seconds (T) after the truck scan is complete.
- 7.1.4 The MEP System data should be available to the MEP Work Center or NII Cloud in one (1) second after the truck scan is complete (O).

7.2 MEP Unit Performance Requirements

- 7.2.1 The MEP Unit shall scan conveyances traveling with speeds of at least 2 to 10 MPH.
- 7.2.2 The MEP Unit shall have a minimum throughput of 80 (T) vehicles per hour.
- 7.2.3 The MEP Unit should have a minimum throughput of 100 (O) vehicles per hour.

7.3 MEP Unit X-Ray System Performance Requirements

- 7.3.1 Low Energy Penetration: The MEP Unit low energy scanning of the cab shall penetrate at least 1.5 inches (T) of steel as measured in accordance with ANSI N42.46-2008.
- 7.3.2 Low Energy Penetration: The MEP Unit low energy scanning of the cab should penetrate at least 2 inches (O) of steel as measured in accordance with ANSI N42.46-2008.

- 7.3.3 High Energy Penetration: The MEP Unit high energy scanning system shall penetrate 9" (T) or greater of steel as measured in accordance with ANSI N42.46-2008.
- 7.3.4 High Energy Penetration: The MEP Unit high energy scanning system should penetrate 12" (O) or greater of steel as measured in accordance with ANSI N42.46-2008.
- 7.3.5 Resolution: The MEP Unit shall have a horizontal/vertical spatial resolution of 0.5 inches (T) or better in accordance with ANSI N42.46-2008.
- 7.3.6 Resolution: The MEP Unit should have a horizontal/vertical spatial resolution of 0.125 inches (O) or better in accordance with ANSI N42.46-2008.
- 7.3.7 Contrast Sensitivity: The MEP Unit shall have a contrast sensitivity of at least 5% (T) at 10%, 50% and 80% of the system's maximum penetration for transmission-based image with the target traveling through the portal at a speed of 5.0 mph
- 7.3.8 Contrast Sensitivity: The MEP Unit should have a contrast sensitivity of at least 2% (O) at 10%, 50% and 80% of the system's maximum penetration for transmission-based image with the target traveling through the portal at a speed of 5.0 mph
- 7.3.9 The MEP Unit shall support display of radiographic images at 4096 (T) gray levels or greater.
- 7.3.10 The MEP Unit should support display of radiographic images at 65,536 (O) gray levels or greater.
- 7.3.11 The MEP X-ray image shall be available to the MEP Work Center or Cloud within five (5) seconds (T) after the truck scan is complete.
- 7.3.12 The MEP X-ray image shall be available to the MEP Work Center or Cloud within three (3) seconds (O) after the truck scan is complete.
- 7.3.13 The system shall display radiographic images with the quality necessary for a system-trained CBP officer to be able to detect 24 test bricks (8" x 5.5" x 1.5", ~1.0 kg) packed together as a single unit within uniform cargo (T).
- 7.3.14 The system should display radiographic images with the quality necessary for a system-trained CBP officer to be able to detect 12 test bricks packed together as a single unit within uniform cargo (O).

Note: "Test bricks" are benign test articles developed by LSS that simulate powdered narcotic tightly packaged in blocks of the dimensions and weight noted.

7.4 Under-Vehicle Back-Scatter System (UVBS) Objective - Performance Requirements

- 7.4.1 If included, the UVBS scan speed shall be equal to or greater than the maximum MEP Unit scan speed.

- 7.4.2 If included, the UVBS shall accommodate dual trailer scanning within a single MEP data package.
- 7.4.3 The UVBS shall display images with the quality necessary for a system-trained CBP officer to be able to detect 8" x 8" test articles (hard plastic rectangular boxes) emplaced on the undercarriage (T).
- 7.4.4 The UVBS should display images with the quality necessary for a system-trained CBP officer to be able to detect 6" x 8" test articles emplaced on the undercarriage (O).

7.5 License Plate Reader (LPR) System Performance Requirements

- 7.5.1 The LPR camera capture rate for images shall be 99.5% (T). Note: "Capture" is defined as recording a human-readable image of the license plate.
- 7.5.2 The LPR camera capture rate for images should be 100% (O).
- 7.5.3 The LPR shall read license plates with 99.0 (T) percent accuracy of the plate alphanumeric identification and the jurisdiction in all lighting conditions (including day, night, and twilight) for all 50 U.S. states, 31 Mexican states and 12 Canadian provinces/territories.

Note: Exclusions for physical obstruction or damage are not allowed.

- 7.5.4 The LPR should read license plates with 99.5 (O) percent accuracy of the plate alphanumeric identification and the jurisdiction in day and night conditions for all 50 U.S. states, 31 Mexican states and 12 Canadian provinces/territories.

Note: Physical obstructions or damage will not be included when calculating accuracy.

7.6 RFID Reader System Performance Requirements (Land Border configuration)

- 7.6.1 The MEP RFID system shall read the DTOPS RFID tags with 99% (T) accuracy for readable tags.
- 7.6.2 The MEP RFID system should read the DTOPS RFID tags with 100% (O) accuracy for readable tags.
- 7.6.3 RFID reads of DTOPS tags in other than the correct lane (i.e. colloquially known as cross-lane reads) shall occur at less than 0.2 (T) percent of all RFID reads.

- Note: Where more than one MEP system is installed, or other RFID readers are installed in proximity to one another.

7.7 Optical Character Recognition (OCR) System Performance Requirements

- 7.7.1 The OCR camera capture rate for images shall be 99.5% (T). Note: "Capture" is defined as recording a human-readable image of the container ID and/or license plate.

- 7.7.2 The OCR system shall read the container number in the seaport environment configuration with 90% (T) accuracy in both day and nighttime conditions.
- 7.7.3 The OCR system should read the container number in the seaport environment configuration with a 95 % (O) accuracy in both day and nighttime conditions.

7.8 Check-In/Staging Lane Kiosk Performance Requirements

- 7.8.1 The kiosk LPR and RFID shall meet the same performance requirements of the MEP Unit LPR and MEP Unit RFID.
- 7.8.2 The QR camera capture rate for images shall be 99.5% (T). Note: "Capture" is defined as recording a machine-readable image of QR coded.
- 7.8.3 The QR camera capture rate for images should be 100% (O).
- 7.8.4 The QR shall read the QR code with 99.0 (T) percent accuracy.
- 7.8.5 The QR should read the QR code with 99.9 (O) percent accuracy.

7.9 CCTV and Camera Systems Performance Requirements

- 7.9.1 Area surveillance component (beyond MEP lanes) shall provide a recording capacity of 30 calendar days for the retrieval of video.

8 Interface Requirements

The interface requirements for the MEP to the NII Control Center and Cloud are described here.

8.1 MEP System Interface Requirements

- 8.1.1 The MEP System shall interface to the NII Control Center, (future) Cloud, and NII-I system in accordance with *Attachment 02, Integration* and the associated Interface Control Document (ICD) of the SOW.
- 8.1.2 The MEP System shall support interface to Government network and (future) Cloud storage of all MEP data elements including image and metadata, health, performance, fault and configuration status, without degradation of existing network performance.
- 8.1.3 The MEP System shall include LPR, RFID, QR and/or OCR metadata to identify a particular vehicle and cargo shipment, and to associate it with its NII image package as the vehicle proceeds through the POE scanning and inspection process.

8.2 MEP X-ray Unit Interface Requirements

- 8.2.1 The System shall interface to any RPMs in proximity over a hardwired connection (e.g. fiber or twisted pair as required) to enable RPM blanking.

8.3 Under-Vehicle Back-Scatter System (UVBS) – Objective Interface Requirements

8.3.1 If included, the UVBS images shall be integrated into the data package.

8.4 Traffic Control System (TCS) Interface Requirements

8.4.1 The TCS traffic lights shall include a red-light representing stop and green light as the indicator to proceed.

8.4.2 Gate arms (if required) shall provide an alarm to the control center MEP operator if they reverse on contact with an obstruction.

8.4.3 The TCS shall include mechanical indicators to warn and divert drivers of oversized vehicles which could damage MEP Unit prior to entering the MEP lane and be placed to allow sufficient maneuvering space for the truck to avoid entering the MEP.

8.4.4 The TCS shall include active signage to direct vehicles to open MEP lanes as well as past the MEP lanes when necessary.

8.5 Vehicle Sensors Interface Requirements

8.5.1 The Vehicle Speed Indicator System shall provide a visual cue to the driver communicating his/her speed and whether they are within the correct speed range for effective scanning.

8.6 Data Storage (DS) Interface Requirements

8.6.1 The MEP System shall export data packages in accordance with *Attachment 02, Integration* and the associated ICD of the SOW.

8.6.2 Locally stored files, records, and scan images shall be viewable on MEP Work Station.

8.6.3 Locally stored files, records, and scan images shall be accessible by administrators who are able to copy to removable media storage devices: encrypted, FIPS 1402 Level 2 or greater and FIPS 197 compliant flash and hard drives.

8.6.4 The MEP System DS shall create a warning when storage reaches 70% of its maximum storage capacity.

8.6.5 The MEP System DS shall create an alert when storage reaches 90% of its maximum storage capacity

8.6.6 The MEP System DS shall create an alarm when storage reaches 99% of its maximum storage capacity.

9 Survivability Requirements

Survivability requirements describe the capability of a system to fulfill its mission in a timely manner in the presence of attacks, failures, or accidents.

9.1 MEP System Survivability Requirements

- 9.1.1 The MEP System shall include hardware to protect from lighting strikes or power surges.
- 9.1.2 The MEP System shall condition incoming power to within $\pm 10\%$.
- 9.1.3 The MEP System shall have backup capability to maintain power sufficient to allow thirty (30) minutes of standard operation and a graceful shut down of systems.
- 9.1.4 The MEP System exterior components and housings shall be high strength, anti-corrosive, and weathertight for the full design life of 10+ years.
- 9.1.5 In the event of an outage, the MEP system shall transmit compliant data to CBP once the system is restored.

9.2 MEP Unit Survivability Requirements

- 9.2.1 The MEP Unit shall have a Power Conditioner and UPS, Online Double Conversion type, to protect the system from electrical power degradation/fluctuations/loss.

9.3 Under-Vehicle Back-Scatter System (UVBS) – Objective Survivability Requirements

- 9.3.1 If included, the UVBS shall have provisions for preventing water intrusion or otherwise manage water which infiltrates the unit to prevent damage/failure of the unit.
- 9.3.2 If included, the UVBS housings shall be high strength, anti-corrosive, and weathertight for the full design life of 10+ years.

9.4 CCTV and Camera Systems Survivability Requirements

- 9.4.1 Backup power shall be provided for all surveillance system components for one (1) hour of operation.

9.5 MEP Work Center Survivability Requirements

- 9.5.1 The system shall include an UPS, Online Double Conversion type, for the MEP Work Center and a power conditioner to regulate and maintain power sufficient to allow thirty (30) minutes of standard operation and a graceful shut down of systems.

9.6 NII Control Center Survivability Requirements

- 9.6.1 The NII Control Center shall be connected to a backup generator system capable of running all direct and support functions of the NII Control Center for a minimum period of 24 hours.

9.6.2 The NII Control Center shall have redundant backup system requirements to include, at a minimum, backup electrical power.

Note: Most LPOE have backup generators that should be able to provide backup power to the command center. Any backup power requirements will be discussed with GSA.

10 Human Systems Integration (HSI) Requirements

HSI requirements include human factors requirements, such as human-machine interface or ergonomic requirements.

10.1 MEP Unit HSI Requirements

10.1.1 The MEP System shall provide audio and visual cues in and around the scan area when the MEP Units are in scanning (X-ray transmit) mode.

10.1.2 The MEP System shall provide audio and visual cues when the MEP Units are in scanning (X-ray transmit) mode

10.2 MEP Work Station HSI Requirements

10.2.1 The MEP System shall provide audio and visual cues to the MEP Work Stations when the MEP Units are in scanning (X-ray transmit) mode

10.3 NII Control Center HSI Requirements

10.3.1 The system Human Machine Interface (HMI) shall turn off X-ray upon operator command.

10.3.2 The MEP System shall provide audio and visual cues in the NII Control Center when the MEP Units are in scanning (X-ray transmit) mode

11 Environmental, Safety, and Occupational Health (ESOH) Requirements

Describe the requirements necessary to comply with existing ESOH policies or regulations (e.g., National Environmental Policy Act [NEPA], Occupational Safety and Health Act [OSHA], Clean Air Act [CAA]) and minimize environmental, safety, and occupational health risks.¹

11.1 MEP System ESOH Requirements

11.1.1 The MEP System shall maintain full operability in the environmental extremes seen on the northern and southern border without degradation in its intended outdoor environment.

11.1.2 The MEP System shall include safety interlocks such as operator proximity/presence sensors for all systems for the protection of pedestrian traffic and untrained personnel.

11.1.3 The MEP System shall operate under environmental conditions along the northern and southern borders as specified in the table below MEP System Environmental Conditions.

Table 4. MEP System Environmental Conditions

Category	Requirement
Relative Humidity	All system components shall be able to operate during and after exposure to relative humidity (RH) levels of up to 98% RH at the range of ambient temperatures stated below.
Moisture and Dust Protection	Comply with IP code 54 (see IEC 60529), the ingress of dust and splashing water.
Heat Loading	All system components shall operate with a maximum heat loading of 110 Watts/square foot due to direct sunlight without performance degradation.
Temperature	Ambient temperature ranges from -40°C to +55°C (-40°F to 131 °F).
Wind	Withstand wind speeds of up to 125 mph.
Environmental Sealing	N42.41-2007 requirements in Section 6.10 (for outdoor use)
Electromagnetic Performance	ANSI N42.41-2007 requirements in Sections 6.3-6.8
Precipitation	Up to 2 inches / hour
Salt Fog	5% by weight. Reference MIL-STD-810, Method 509.4

11.1.4 The MEP System shall include individual Emergency Stop Switches for all systems in the control center at the MEP operator position and on the MEP systems themselves, outside of the Radiation Control Area (RCA) to manually interrupt the scanning process by disabling the transmission of X-rays in an emergency or when there is an unauthorized entry inside the system's RCA in accordance with ANSI N43.3-2008, paragraph 7.05.2, and ANSI N43.14-2011.

11.1.5 The MEP System shall include devices (bollards, guard rails, etc.) to prevent/protect the system and associated subsystem/components from being hit by vehicles traveling through/near the system.

11.1.6 The MEP System shall have guard rails that ensure all conveyances (vehicles and trailers) with an outer track width (i.e. the dimension from the outer tire tread on one tire to the outer tire tread on the opposite tire on the same end of the

vehicle) equal to or less than the width of the scanning area are able to pass through the system with clearance of 6" (i.e. if the scanning area is 8' 6" wide, then guard rails are at least 9' wide).

11.2 MEP Unit X-Ray System ESOH Requirements

- 11.2.1 The MEP Unit shall not rely on radioactive isotopes.
- 11.2.2 The MEP Unit shall perform X-ray transmission and/or backscatter scan of the tractor/cab of the truck at low energy/flux allowing a safe exposure rate.
- 11.2.3 The radiation dose to vehicle occupants in adjacent lanes shall not exceed 50 μ R in any one hour.
- 11.2.4 The MEP Unit shall have a controlled RCA which limits the potential exposure to personnel outside of this area to < 50 μ R in any one hour.
- 11.2.5 Engineering controls such as barricades or fencing shall be installed to maintain the controlled RCA
- 11.2.6 The MEP shall allow subject vehicle speeds of less than two (2) miles per hour, however, if scanning at such a speed would expose vehicle occupants to more than 25uR per scan, then the MEP shall stop scanning
- 11.2.7 The MEP System, plus UVBS if equipped, shall limit radiological dose equivalent to each truck driver or passenger being scanned to no more than 25 μ R/scan (T).
 - Note: Requirement must be met at >2 mph and <10 mph scan speed and analysis include adjacent lane vehicles and impact of traffic queue delays on MEP radiation exposure to drivers or passengers, per ANSI 43.17 2009.
- 11.2.8 The MEP System, plus UVBS if equipped, should limit radiological dose equivalent to each truck driver or passenger being scanned to no more than 10 μ R /scan (O).
 - Note: Requirement must be met at ≥ 2 mph and ≤ 10 mph scan speed and analysis includes adjacent lane vehicles and impact of traffic queue delays on MEP radiation exposure to drivers or passengers, per ANSI 43.17 2009.
- 11.2.9 Approval from the CBP Radiation Safety Officer (RSO) shall be obtained for installations that do not have X-ray opt-out lane(s).

11.3 Traffic Control System (TCS) ESOH Requirements

- 11.3.1 A hard barrier capable of withstanding a low-speed strike (≤ 15 mph) shall be placed immediately prior to the entrance of the MEP as a safeguard.

11.4 CCTV and Camera Systems ESOH Requirements

- 11.4.1 The CCTV System shall provide operators with full situational awareness for the MEP System and adjacent operations.

12 Interoperability Requirements

Interoperability requirements allow the MEP System to operate without impacting other systems or components.

12.1 MEP System Interoperability Requirements

12.1.1 The MEP System shall be able to operate independently of all other lanes.

12.2 MEP Unit X-Ray System Interoperability Requirements

12.2.1 The System shall not interfere with RPM gamma-ray detection operations.

12.2.2 The System shall employ a pulsed photon generator or other similar technology (e.g. linac, betatron or similar) to facilitate RPM blanking.

12.2.3 The System shall produce an RS422 (or RS485) compatible blanking signal with adjustable timing which can be transmitted to any RPMs in proximity over a hardwired connection (e.g. fiber or twisted pair as required).

12.2.4 The blanking signal shall remain active regardless of MEP operational status.

12.2.5 The system shall be capable of meeting the requirements in this standard while operating at a fixed photon pulse frequency of 200 Hz, 400 Hz, or 800 Hz to integrate with existing blanking infrastructure.

12.2.6 If multiple NII systems are deployed in close proximity, the photon pulse frequency from each system shall be synchronized and produce a single blanking signal to interface with RPMs in proximity.

12.2.7 The System shall not interfere with RPM neutron detection operations.

12.2.8 Any neutron blanking signal should be delivered independently of the gamma blanking signal to minimize the neutron blanking duration (O).

12.2.9 Systems shall be designed and shielded to reduce the production of neutrons to the level of background at any RPMs in proximity (T).

12.2.10 Systems should be designed and shielded to reduce the production of neutrons to the level of background at 100 feet in all directions from the system center (O).

12.3 Under-Vehicle Back-Scatter System (UVBS) Objective - Interoperability Requirements

12.3.1 If included, the UVBS shall be fully integrated with and use the MEP Unit LPR system.

12.4 Vehicle Sensors Interoperability Requirements

12.4.1 Each MEP Unit 'lane' vehicle presence sensor shall be integrated with the traffic control system.

12.5 CCTV and Camera Systems Interoperability Requirements

12.5.1 The Fixed CCTV system shall be independent of the PTZ cameras at the entrance and exit of the MEP Units.

12.6 MEP Work Center Interoperability Requirements

12.6.1 Each MEP Work Center shall allow any analyst to adjudicate any X-ray image from any of the connected MEP Units.

13 Security & Privacy Requirements

This section describes requirements to protect data and information (e.g., screening images) from intrusion, corruption, repudiation, theft, unauthorized access and control of access to the data. This includes controlling who may view and alter application data.

Note that Section 13 uses a different set of requirement groups (not component number). These requirement groups are more appropriate to the Security and Privacy topic than component numbers.

13.1 MEP System Security & Privacy Requirements

13.1.1 Audit Record Retention – Audit records shall be maintained on the system for a minimum of ninety (90) days.

13.1.2 Audit Record Retention - Audit records shall be preserved for a period of 7 years as part of managing records for each system to allow audit information to be placed online for analysis with reasonable ease.

13.1.3 Audit Reduction/Report Generation – The system shall provide an audit reduction and report generation capability that: 1) Supports on-demand audit review, analysis, and reporting requirements and after-the-fact investigations of security incidents, and 2) without altering the original content or time ordering of audit records.

Note: DHS 4300A policy requires audit record generation capability for relevant or auditable events identified by DHS. National Institute of Standards and Technology (NIST) defines a relevant event as an occurrence (e.g., an auditable event or flag) considered to have potential security implications to the system or its environment that may require action.

13.1.4 Data Storage (DS) Security & Privacy Requirements – The MEP System DS shall store data in encrypted form.

13.2 MEP Work Center Security & Privacy Requirements

13.2.1 The MEP workstations shall be compliant with the latest CBP Technical Reference Model (TRM) specification.

13.3 Access Control Security & Privacy Requirements

- 13.3.1 Unique User Accounts – The system shall enable all users to have unique account identifiers, including separate identifiers for operator accounts and administrator accounts.
- 13.3.2 Automatic Account Lockout – The system shall lock a user account for twenty (20) minutes after three consecutive failed logon attempts
- 13.3.3 Automatic Session Termination – The system shall lock the user login session after twenty (20) minutes of inactivity.
- 13.3.4 Automatic Session Termination – The system shall require the user to authenticate with the user account password to regain system access after the system lock.
- 13.3.5 Automatic Session Termination – After 60 minutes of inactivity, the system shall log the user account out of the system.
- 13.3.6 Warning Banner – The DHS Chief Information Security Officer (CISO) stipulates that a warning banner statement be displayed on all DHS systems during user account logon. The system shall prompt the user to accept the current language before they can authenticate with the user id and user password to access the system. The current language required follows: 1. You are accessing a U.S. Government information system, which includes (1) this computer, (2) this computer network, (3) all computers connected to this network and (4) all devices and storage media attached to this network or to a computer on this network. This information system is provided for U.S. Government-authorized use only. 2. Unauthorized or improper use or access of this system may result in disciplinary action, as well as civil and criminal penalties. 3. By using this information system, you understand and consent to the following: a. You have no reasonable expectation of privacy when you use this information system; this includes any communication or data transiting, stored on, originated from or directed to this information system. At any time, and for any lawful Government purpose, the Government may monitor, intercept, search and seize any communication or data transiting, stored on, originating from or directed to or from this information system. b. The Government may disclose or use any communications or data transiting stored on, originated from or directed to or from this information system for any lawful Government purpose. c. You are NOT authorized to process classified information on this information system.

13.4 Auditing Security & Privacy Requirements

- 13.4.1 The System shall retain audit records with sufficient detail to facilitate the reconstruction of events if a compromise or malfunction occurs.
- 13.4.2 Audit records shall contain at least the following information: 1. Identity of each user and device accessing or attempting to access the system. 2. Time and date of the access and the logoff. 3. Activities that might modify, bypass, or negate

information security safeguards. 4. Security – relevant actions associated with processing. 5. All activities performed using an administrator account/identity.

13.4.3 Audit Record Access – Audit records and audit logs shall be protected from unauthorized access, modification, or destruction.

13.5 Computer Security & Privacy Requirements

13.5.1 System security shall follow DHS/CBP Enterprise and Security Architecture requirements for a system categorized as Moderate/Low under FIPS-199 and collecting privacy sensitive information.

13.5.2 The System shall be compliant with the latest version of the DHS Sensitive Systems Policy Directive 4300A, (i.e. Version 13.2 as of August 2025).

13.5.3 The System shall be compliant with CBP Information Systems Security Policies and Procedures, HB 1400-05D, Version 9.0 (or latest) for access control.

13.5.4 The system HMI shall disable wireless communications, upon administrator command.

13.5.5 The System shall comply with the DHS hardening guides for operating systems.

13.5.6 The System shall comply with the DHS configuration guides for applications.

13.5.7 The System DS shall be encrypted in accordance with FIPS 197 Advanced Encryption Standard (AES), Chapter 5, algorithms.

13.5.8 The System shall encrypt with at least 256 bit in accordance with FIPS 140-2 Standards for Security Requirements for Cryptographic Modules.

13.5.9 Desktop computers shall be equipped with operating systems of Windows 11 or greater that are on the CBP TRM.

13.5.10 Basic Requirements - Any legacy air gapped non-integrated systems shall conform with the following requirements:

- Use upgradable operating system software and approved applications,
- Not to use or disable Wi-Fi,
- Use operating systems that meet the requisite OHS STIGs for hardening configurations,
- Use required encryption for data at rest,
- Be cyber resilient to data manipulation, unauthorized download and non-malicious incidents.

13.5.11 Integrated networked systems shall follow the requirements of non-integrated systems in addition to:

- Only use WI-FI designs which are explicitly reviewed and approved by the CBP Enterprise Infrastructure and Operations Directorate (EIOD),
- Only use other wireless connectivity as approved by EIOD,
- Be patched and maintained according to OHS patching standards (within 30 days usually),
- Not allow Username/Password authentication,
- Require PIV authentication,
- Integrate with CBP identify management lifecycle systems,
- Integrate with enterprise audit log repository (SPLUNK),
- Be capable of receiving operating system security patches and updates and leverage CBP enterprise security notification services for network security alert monitoring.

13.6 Identification and Authentication Security & Privacy Requirements

13.6.1 Levels of Access – The System shall provide two levels of CBP User access (Operator and Administrator).

13.6.2 Administrators shall be able to perform functions described in 5.13.2, 8.6.3, and 13.5.4.

13.6.3 MEPs not integrated/connected with the CBP network shall comply with the following. Passwords – All passwords used to access the NII system for operation shall be required to be Strong Passwords. Strong passwords have the following requirements: 1) Are at least 12 characters in length. 2) Comply with the DHS hardening guides for operating systems and the configuration guides for applications. In the absence of guidance, the ISSO will determine the appropriate password complexity based on the level of risk. 3) The system shall force all passwords to contain three of the four following types of characters. i. Uppercase Alphabetic characters. ii. Lowercase Alphabetic characters. iii. Special characters. iv. System shall check that the passwords are not the same as any of the user's previous 8 passwords when creating passwords. If they are the same as any of the user's previous 8 passwords the system shall force the user to change to a password that is not the same as any of the previous eight passwords. v. Strong passwords are not the same as the User identification (ID). vi. Strong passwords shall contain special characters. vii. Strong pass phrases, if used in addition to or instead of passwords, are forced to follow the same guidelines as above.

13.6.4 The NII system must have the following capabilities: 1) The system shall deactivate unused accounts automatically if they have not been used in 45 days. 2) Passwords shall not be set to the null string. 3) Automatic passwords shall change frequently. The system requires the user to change their password every 90 days. 4) Password Storage – The system shall store all user account passwords in encrypted form in accordance with DHS/CBP encryption requirements.

13.7 Printing Security & Privacy Requirements

13.7.1 The System documents shall be printed with "FOR OFFICIAL USE ONLY - LAW ENFORCEMENT SENSITIVE" in the header or footer.

13.7.2 The footer of printed documents shall indicate system type and location.

14 Reliability, Availability, and Maintainability (RAM)

The contractor shall design, engineer, manufacture, and deploy the system and sub-systems to achieve established outcomes and ensure the availability of the capability throughout its intended service life expectancy by applying the tenets of CBP's Cost Wise Readiness (CWR) metrics. Accordingly, the Contractor shall sustain the system and/or key subsystems during the warranty period in a manner that meets the Operational Availability (A_o) service requirements. The Contractor is encouraged to conduct an analytical assessment that offers an alternative balanced solution for optimizing reliability and downtime to achieve readiness.

The Contractor will address the Reliability, Availability, and Maintainability (RAM) in the Lifecycle Sustainment and Support Requirements attachment within the SOW.

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